ENCN205

Applied Data Analysis for Civil and Natural Systems

LECTURE 1

Why Location Matters — Introduction to GIS

Slides and interactive demos (click or scan):
aardid.github.io/uc_205_gis_course

University of Canterbury | Te Whare Wānanga o Waitaha

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"Where is it?" is one of the most powerful question in engineering.

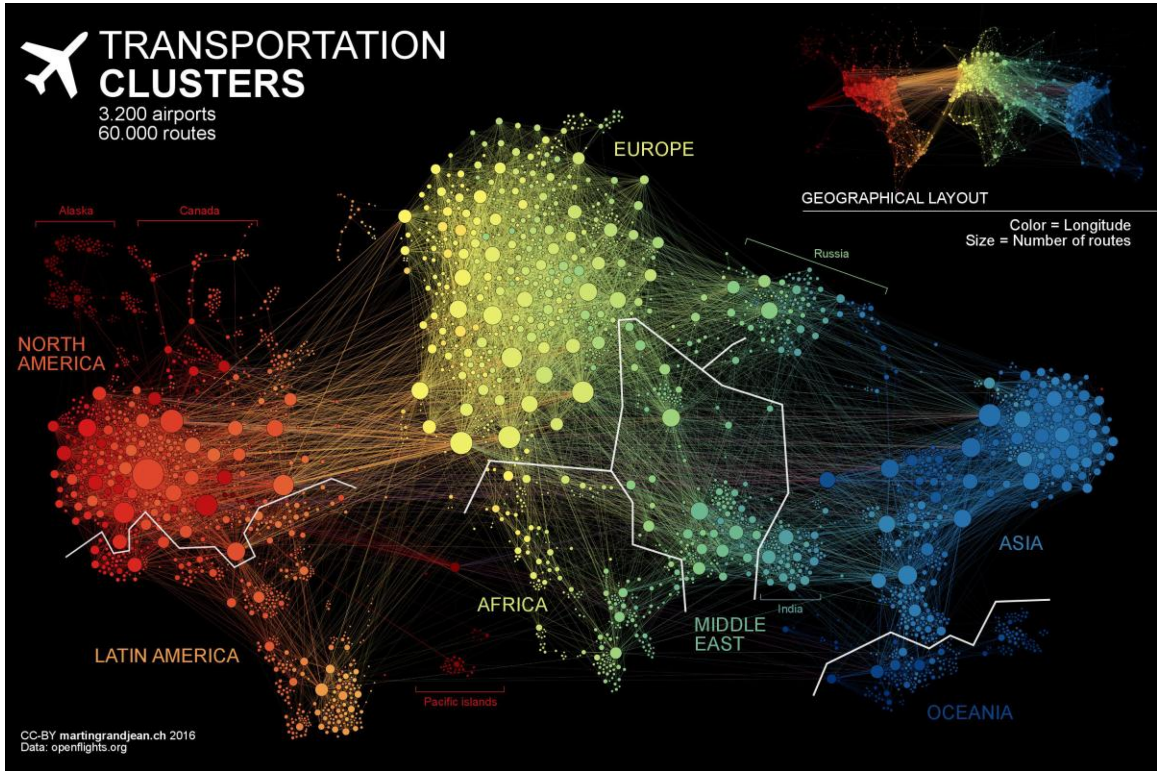
Where should we build the new water intake?

Which buildings will flood if the river rises 2 metres?

Where are the weakest links in the pipe network?

Which communities are most vulnerable to earthquake damage?

3,200 airports and 60,000 routes - spatial data reveals patterns invisible in spreadsheets.



What is spatial thinking?

Spatial thinking is the process of **problem solving** via the **coordinated use of space, representation, and reasoning**. Spatial thinking uses representations to help us remember, understand, reason, and communicate about **the properties of and relations between objects represented in space**, whether or not those objects themselves are inherently spatial.

National Academies of Sciences, Engineering, and Medicine. 2006. Learning to Think Spatially. Washington, DC: The National Academies Press. https://doi.org/10.17226/11019.

The First GIS Analysis

London, 1854 — a cholera outbreak was killing hundreds; , and the prevailing theory says the disease spreads through bad air.

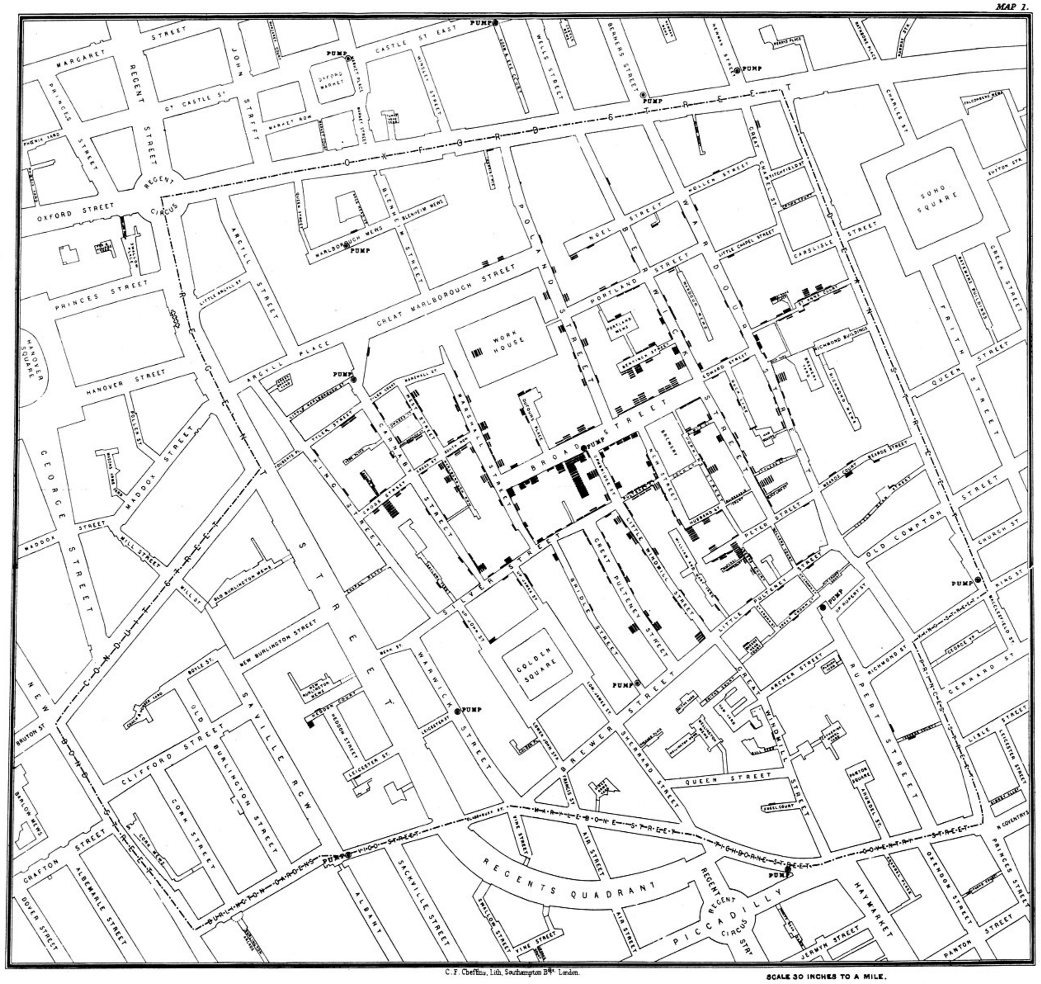
Dr John Snow plotted each death as a dot on a street map.

The dots clustered around one location: the Broad Street pump.

Snow convinced authorities to remove the pump handle.

The epidemic subsided — spatial analysis saved lives.

A dot map solved a public health crisis 170 years ago — the same logic powers modern GIS.



Snow's map didn't show the contaminated sewage source; he still needed the mechanism to prove his case.
The map pointed; the mechanism convinced.

Always ask: '

t showing?'

<ins>**Saved fictional people from fictional zombies**</ins>

<ins>**Saved real people from a real cholera outbreak**</ins>

<ins>**A map is evidence, not proof.**</ins>

Location Carries History: The Alabama Black Belt

A coastline from 100 million years ago is still visible in modern election maps.

Cretaceous seas deposited chalk in a crescent across Alabama

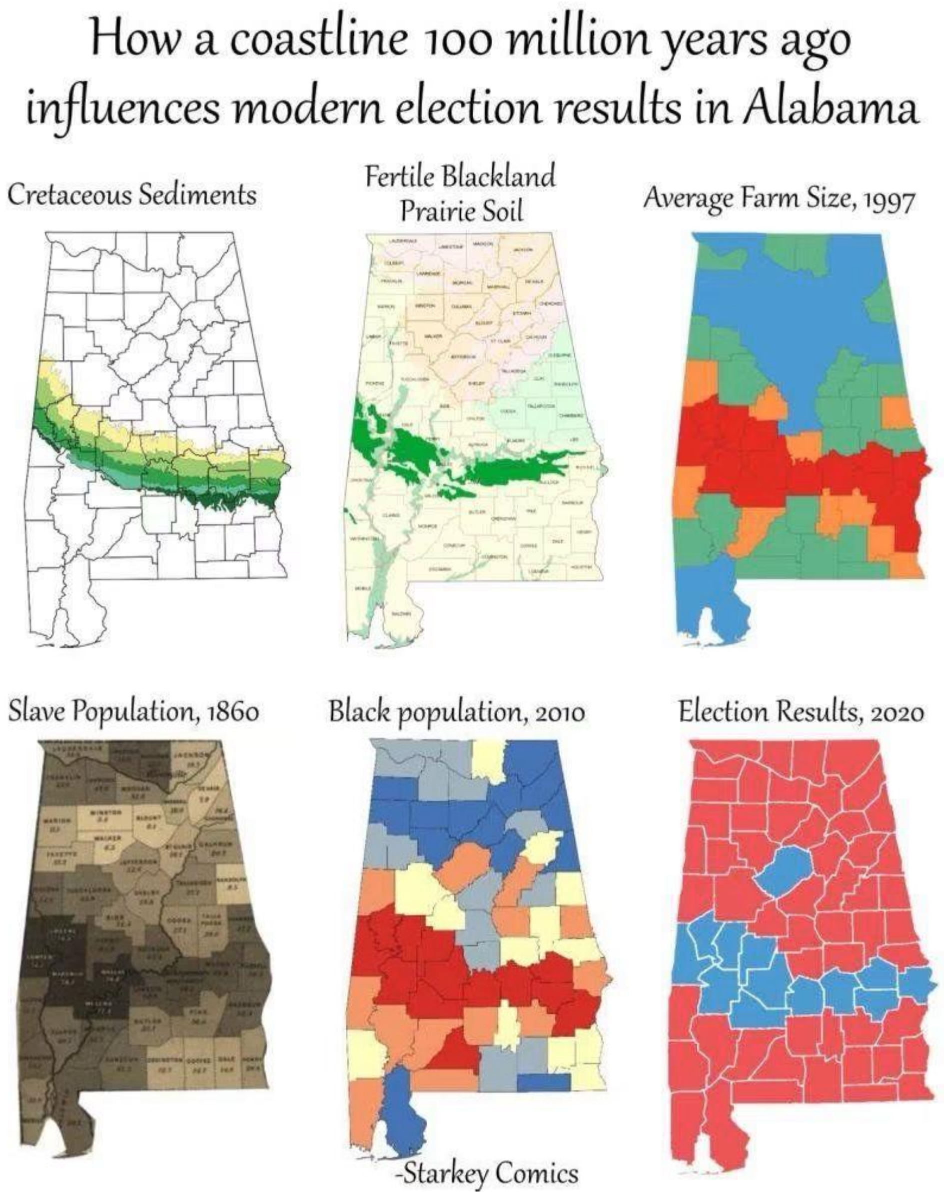
Chalk weathered into fertile, dark 'Blackland Prairie' soil

1800s: fertile soil → cotton plantations → enslaved population concentrated in the crescent

Today: Black communities remain concentrated there — and the 2020 election map traces the ancient shoreline

Geology shapes soils, soils shape land use, land use shapes society - spatial patterns persist for centuries.

The point for you as engineers: spatial patterns persist, sometimes for centuries, and layers of spatial data let us connect causes and effects across time.



Source: Starkey Comics

What is GIS?

A Geographic Information System (GIS) is a framework for:

Collecting spatial and attribute data

Storing and managing geographic datasets

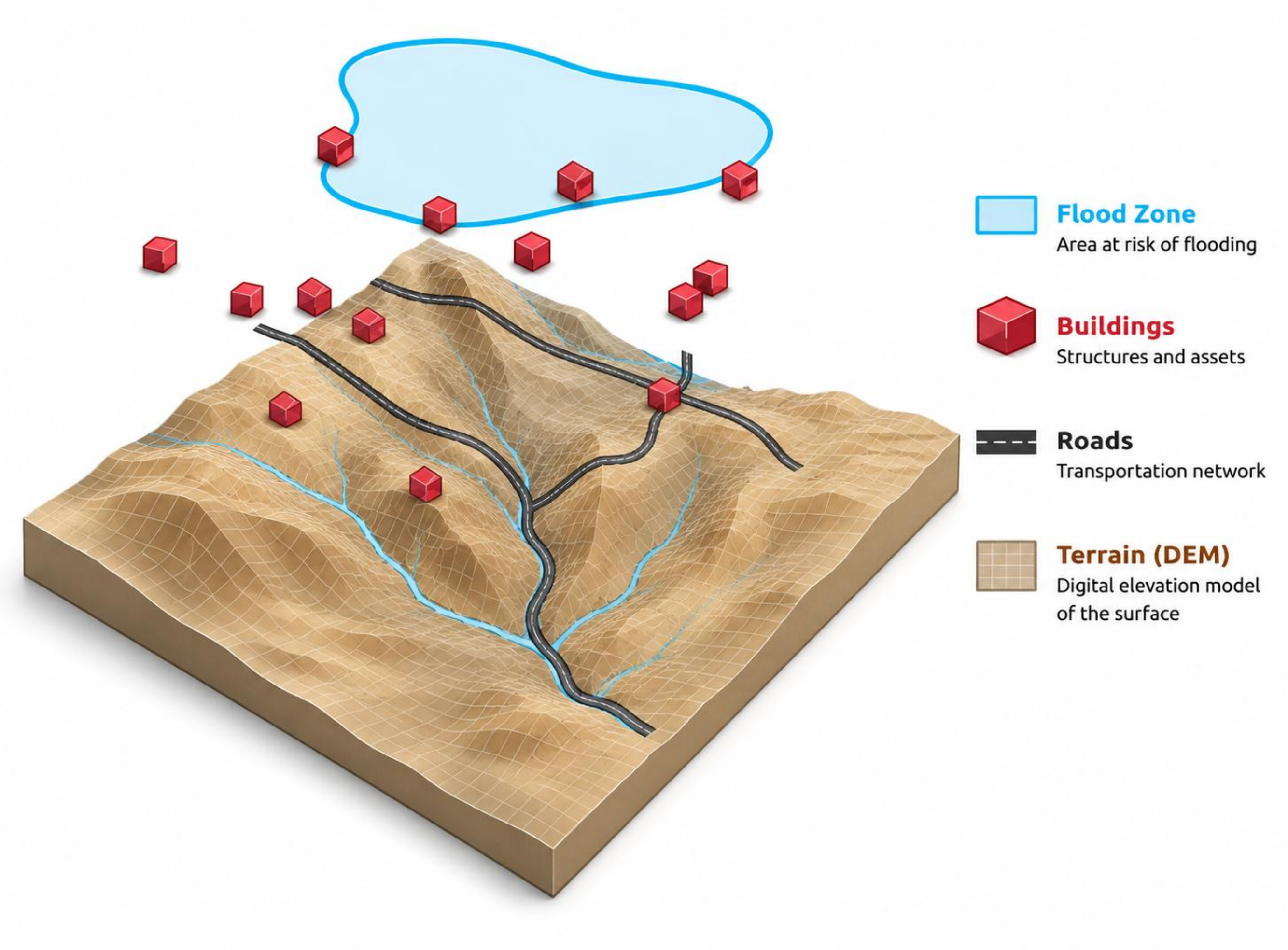
Analysing spatial relationships and patterns

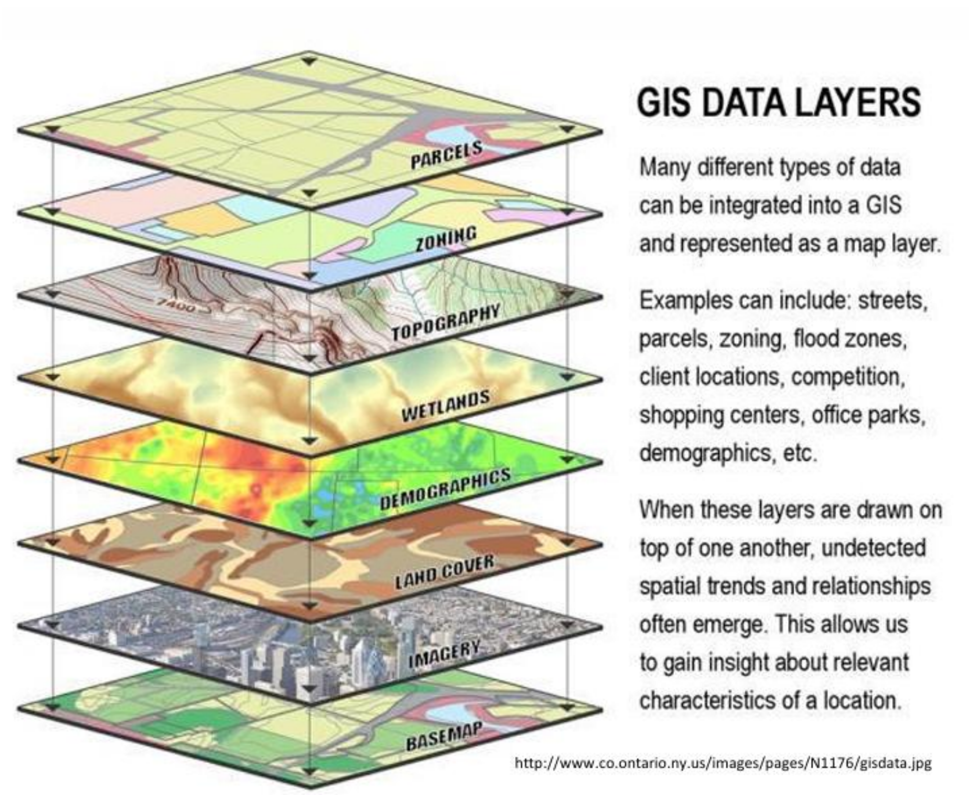
Visualising results as maps and reports

It is both a tool (software) and a science (spatial thinking: knowing which question to ask and which analysis answers it).

Software versions come and go; the thinking is what makes you valuable as an engineer.
(that's what I actually want you to leave this course with)

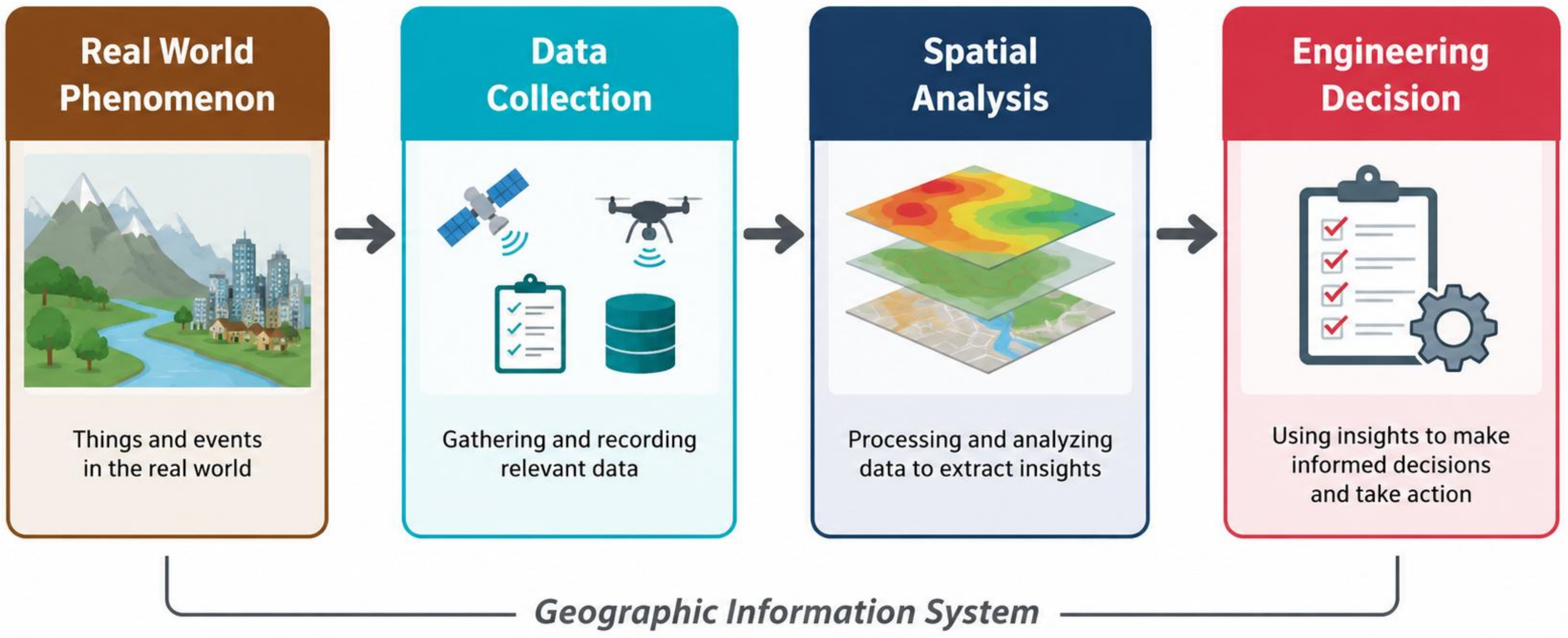
GIS: Layers of Spatial Information





Each layer represents a different aspect of the real world — GIS combines them to answer engineering questions.
Layers plus spatial relationships equals insight.

The GIS Workflow *(how does GIS actually fit into engineering work?)*



Key principle: GIS connects real-world observations to engineering decisions.

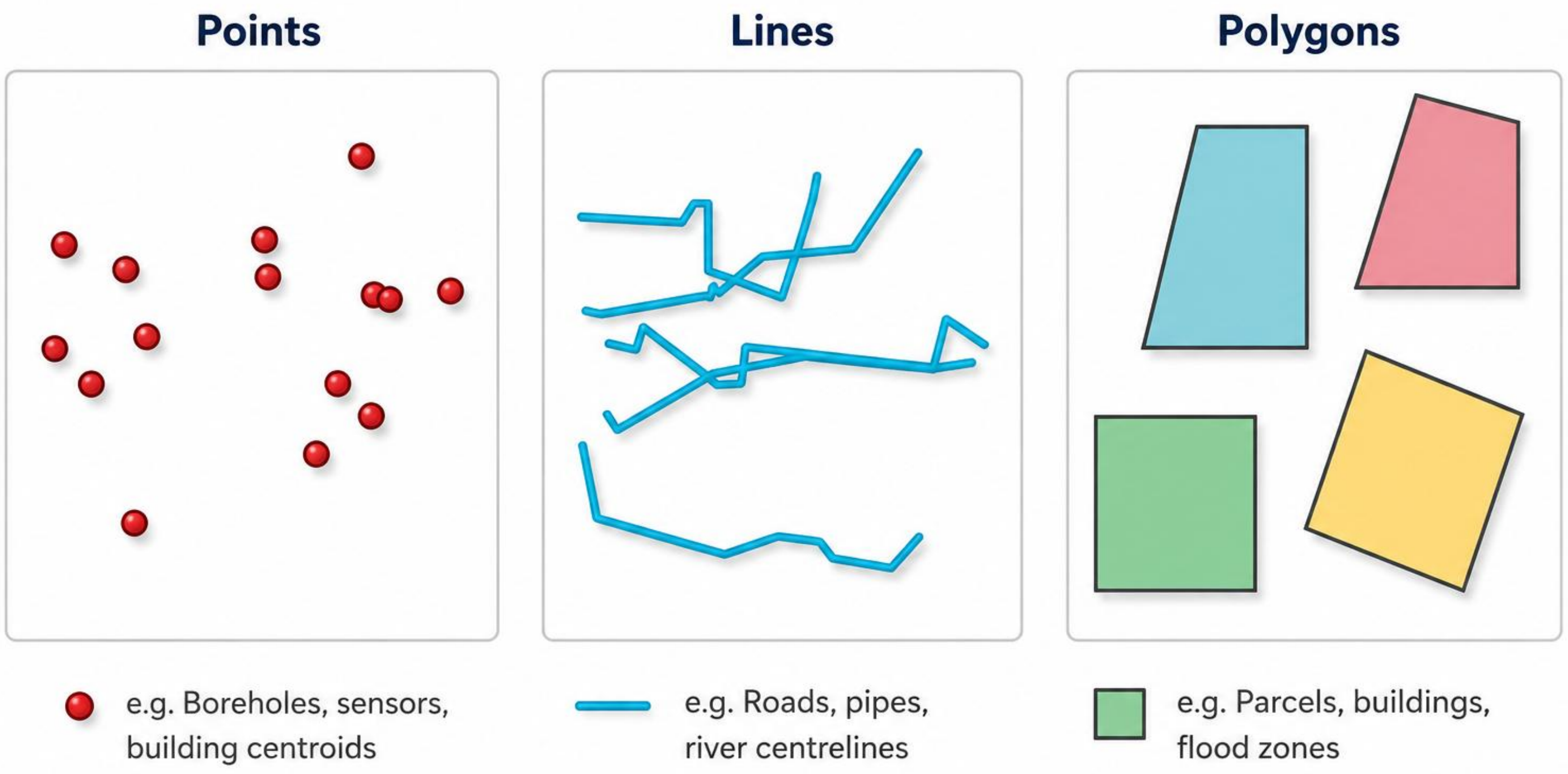
The quality of the decision depends on the quality of every step.

This course teaches you to execute each step using Python.

Always ask: 'what question is this map answering?'

(how do we actually represent the world in a computer?)

Vector Data Model

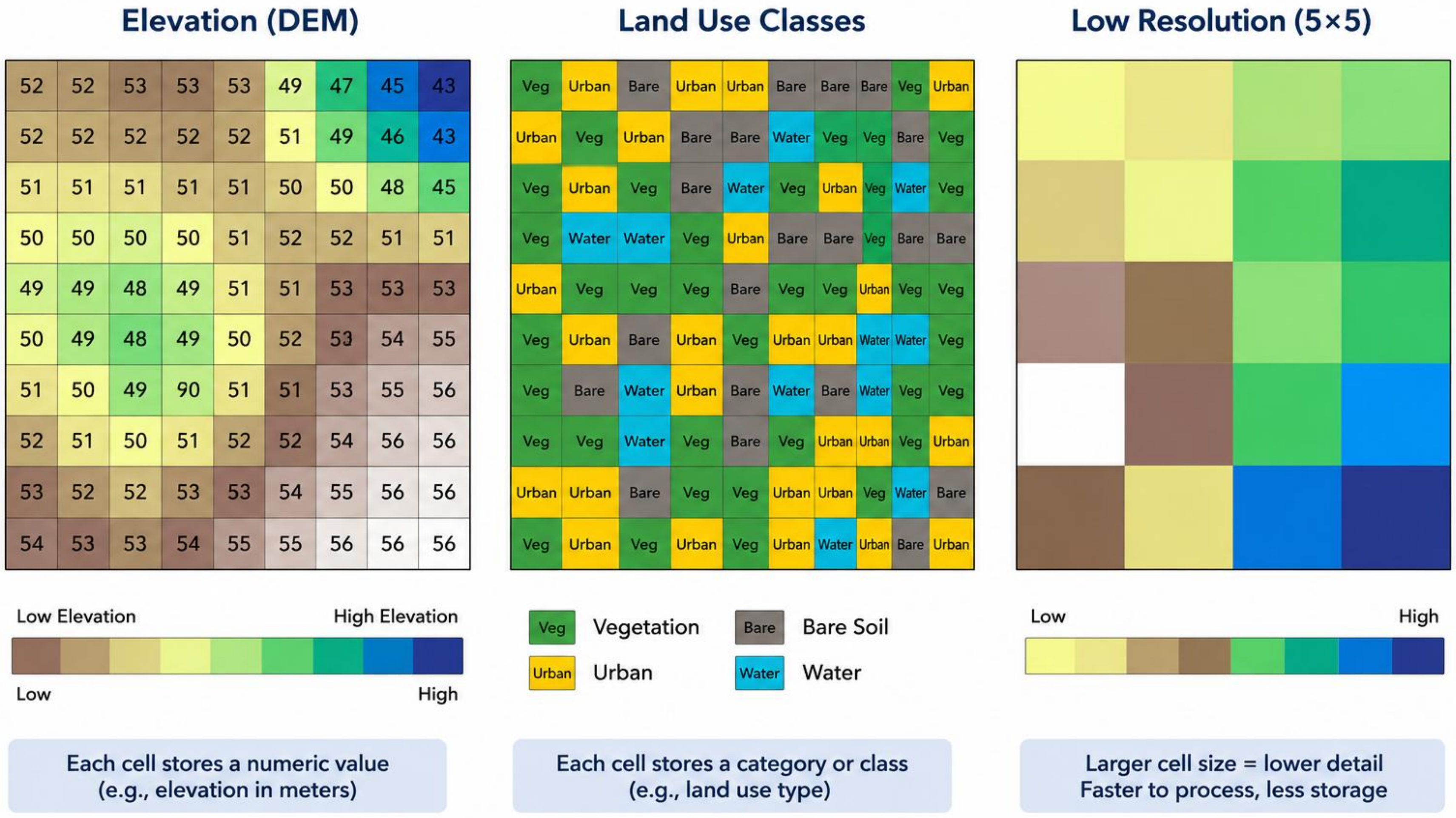


Represents discrete features with exact geometry + attributes

Each feature = a row in a table with a geometry column (GeoDataFrame)

(how do we actually represent the world in a computer?)

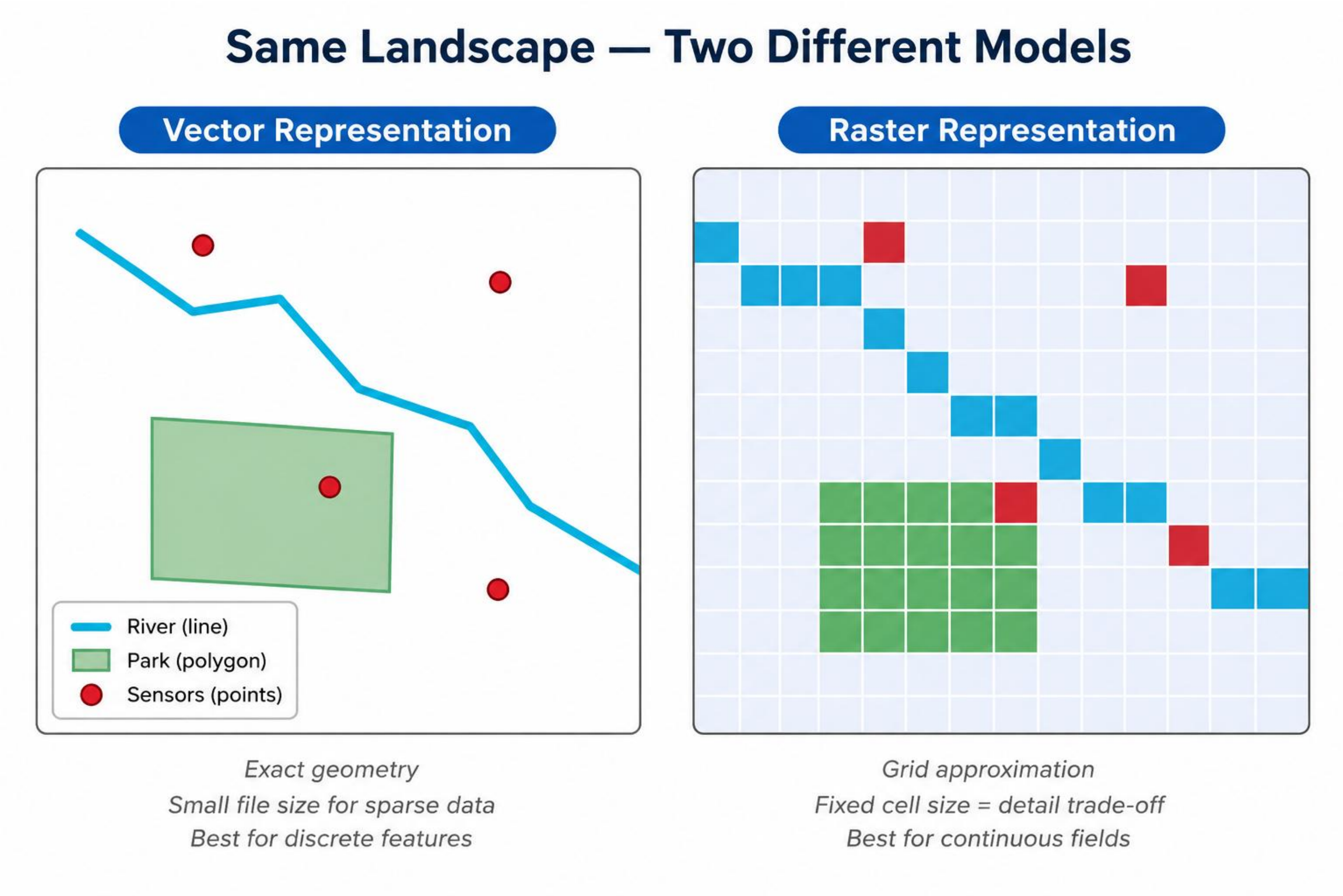
Raster Data Model - Grids of Cells with Values



Represents continuous fields as a grid of cells (pixels), each with a value

Resolution = cell size — determines detail vs. file size trade-off

Vector vs Raster



Choosing the right model depends on: what you're mapping + what analysis you need

LIVE DEMO

L01_LayerExplorer.html

Toggle spatial layers on/off to see how combining data reveals engineering insights that no single dataset can provide.

Scan to open the demo on your phone

aardid.github.io/uc_205_gis_course/L01_LayerExplorer.html

All slides & demos: aardid.github.io/uc_205_gis_course

Data Sources for NZ Engineering

National datasets:

LINZ Data Service — parcels, topo, imagery

Stats NZ — census, deprivation index

ESNZ (former GNS Science) — earthquake, landslide hazards

ESNZ (former NIWA) — climate, hydrology, flood models

OpenStreetMap — roads, buildings, amenities

Canterbury-specific:

Canterbury Maps — regional council portal

CCC Open Data — Christchurch city assets

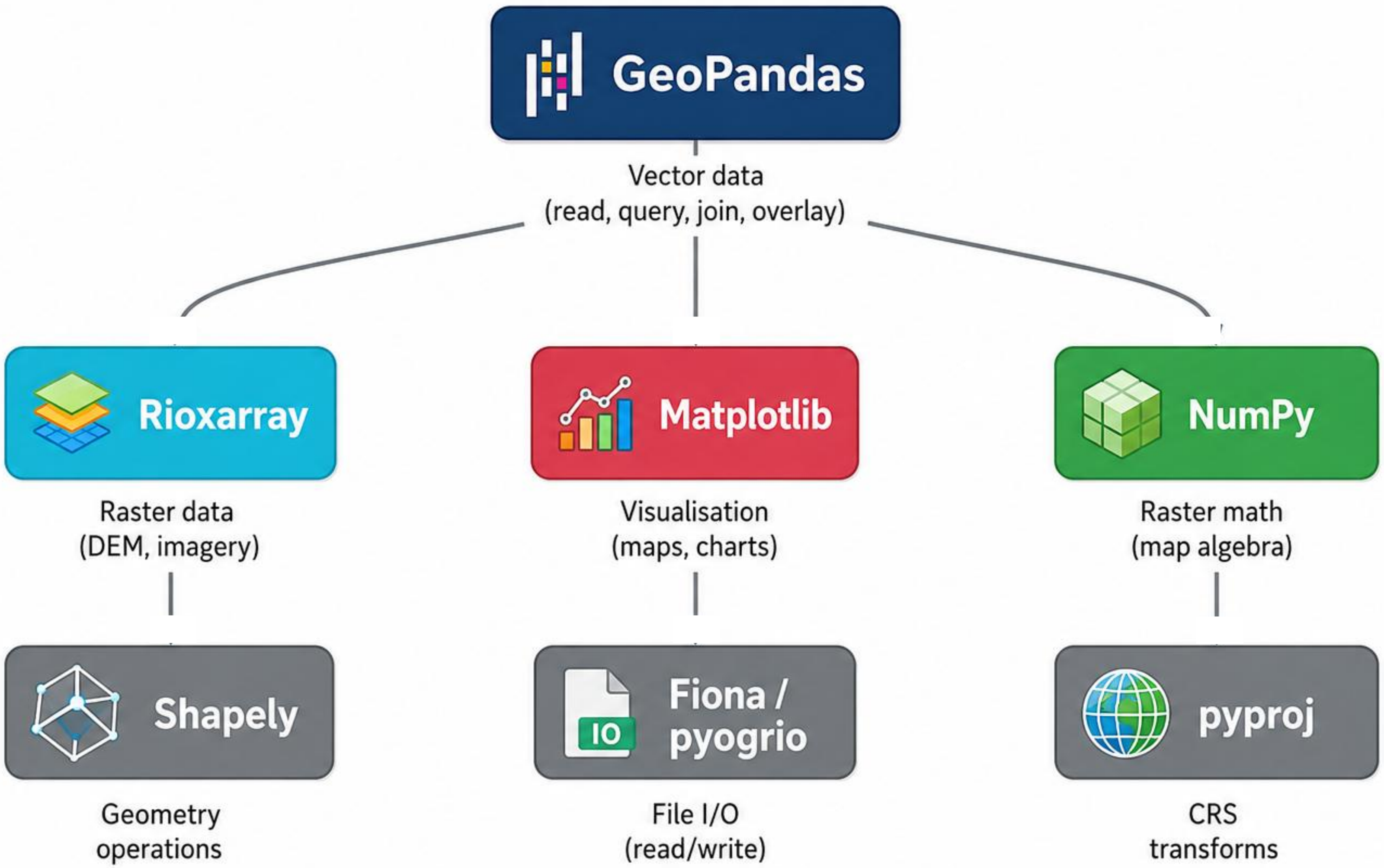
ECan — groundwater, land use, consents

Canterbury Geotechnical Database

All freely available — no licenses needed for this course.

Always ask: 'who collected this - and when?'

The Python Geospatial Ecosystem



All open-source, all free — the same tools used by professional engineers and researchers.

Why Python for Geospatial Engineering?

Reproducible — scripts document every step, unlike clicking through GIS menus

Scalable — process 1 file or 10,000 with the same code

Integrates with statistics — same language for spatial + statistical analysis

Industry standard — used at BECA, Tonkin + Taylor, AECOM, councils

Free — no license fees

You already know Python from term 3 - now we add the spatial dimension.

ENCN205 GIS component – 6 Week Roadmap

Week 7
Module 1
Why location matters: introduction to GIS + guest lecture

Week 8
Module 2
Data quality, ethics & spatial data models

Lab 0 (online)
Introduction to Jupyter notebooks & Python/GeoPandas setup

Week 9
Module 3
Coordinate reference systems & spatial relationships

Lab 1 (Python)
Spatial data basics, CRS & reprojection

Week 10
Module 4
Vector analysis & cartography

Lab 2 (Python)
Vector analysis for engineering exposure

Week 11
Module 5
Raster & hydrological analysis

Lab 3 (Python)
Raster terrain analysis & hazard surfaces

Week 12
Module 6
Risk assessment & the future of GIS

Lab 4 (Python)
Hydrological analysis – Buller catchment

Assessment	Description	Weight (%)	Timing	Learning Outcomes
Lab outputs & guest lecture attendance	Completion of weekly labs (Lab 0–Lab 4) and guest lecture attendance	5	Weeks 7–11	7–10
GIS Final Exam	Comprehensive final exam covering all GIS modules	30	To be announced	7–10

Assessment

Weekly Sign-Off labs - 5%

8 MCQ, pass mark 8/10, unlimited retries

Covers that week's lab

Download a sign-off PNG and upload to Learn

Written Exam - 30%

2 hours, 6 questions, pen and paper

Each question: 10 marks (60 total).

Study tool available - 350 practice MCQs with instant feedback

KEY CONCEPTS

Today's Takeaways

1. GIS is the framework that connects spatial data to engineering decisions
2. Vector = discrete features (points, lines, polygons) with attributes
3. Raster = continuous grids of cell values (elevation, rainfall, land use)
4. Choosing the right data model depends on what you're mapping
5. Python's geospatial ecosystem (GeoPandas + friends) gives you reproducible, scalable GIS

NEXT LECTURE

Lecture 2: Data Quality, Ethics & Professional Practice

By the End of This Course, You Will Be Able To:

- ✓ Load, transform, and query spatial data in Python
- ✓ Choose and apply appropriate CRS for NZ engineering
- ✓ Perform vector analysis: buffers, spatial joins, overlays
- ✓ Perform raster analysis: terrain, flood modelling, map algebra
- ✓ Design clear, professional maps following cartographic principles
- ✓ Delineate catchments and model hydrological processes
- ✓ Conduct multi-criteria site evaluation for engineering decisions

Choosing Vector or Raster

Use Vector when:

Features have clear boundaries

You need exact geometry (lengths, areas)

Data is sparse (not every location has a value)

Examples: buildings, pipes, roads, parcels, zone boundaries

Use Raster when:

The phenomenon varies continuously

You need cell-based maths (map algebra)

Data comes from remote sensing or models

Examples: elevation (DEM), temperature, rainfall, satellite imagery

Discussion

Think of an engineering project you've seen or worked on. What spatial data would you need? Vector, raster, or both?

Examples to consider:

Designing a new subdivision — what layers do you need?

Assessing flood risk for a bridge — vector, raster, or both?

Planning a cycle path network across the city